

Practical 3: Google BigQuery

Q1: WHERE Clause

Untitled...ery

*Practic... nal

+

Practical3 Final

Run

Share

Schedule

Open in

More

Save query

Download

```
3
4 --Q1 WHERE Clause
5 SELECT *
6 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`
7 WHERE EXTRACT(YEAR FROM DATE(Date)) = 2023;
8
```

✓ This script will process 342.34 KB when run.

Using on-demand processing quota

Query results

Chat Save results Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

Results per page: 50 1 – 50 of 998

Q2. Filtering + Conditions

Untitled...ery X *Practic... nal X

Practical3 Final Run Share Schedule Open in More Save query Download

```
8
9 --Q2
10 SELECT *
11 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`
12 WHERE `Total Amount` > (select AVG(`Total Amount`) FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`);
13
```

✓ This script will process 342.34 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
2	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
3	128	2023-07-05	CUST128	Male	25	Beauty	1	500	500
4	220	2023-03-03	CUST220	Male	64	Beauty	1	500	500
5	238	2023-01-17	CUST238	Female	39	Beauty	1	500	500
6	364	2023-08-23	CUST364	Female	19	Beauty	1	500	500
7	408	2023-04-15	CUST408	Female	64	Beauty	1	500	500
8	537	2023-06-03	CUST537	Female	21	Beauty	1	500	500
9	539	2023-06-08	CUST539	Male	25	Beauty	1	500	500
10	541	2023-07-29	CUST541	Male	56	Beauty	1	500	500
11	760	2023-03-27	CUST760	Male	27	Beauty	1	500	500
12	781	2023-12-23	CUST781	Male	35	Beauty	1	500	500
13	934	2023-07-25	CUST934	Male	30	Beauty	1	500	500
14	937	2023-10-23	CUST937	Female	62	Beauty	1	500	500
15	129	2023-04-23	CUST129	Female	21	Beauty	2	300	600
16	131	2023-09-18	CUST131	Female	21	Beauty	2	300	600
17	221	2023-05-07	CUST221	Male	39	Beauty	2	300	600
18	233	2023-12-29	CUST233	Female	51	Beauty	2	300	600
19	290	2023-10-04	CUST290	Female	30	Beauty	2	300	600
20	302	2023-07-14	CUST302	Male	57	Beauty	2	300	600

Results per page: 50 1 – 50 of 350

Q 3. Aggregate functions

The screenshot shows the Google Cloud BigQuery console interface. At the top, the query editor displays the following SQL code:

```
13  
14 --Q3  
15 SELECT SUM(`Total Amount`) AS total_revenue  
16 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`;  
17
```

Below the query editor, a status bar indicates "Using on-demand processing quota". The "Query results" section is active, showing a table with one row of data:

Row	total_revenue
1	456000

The interface includes standard BigQuery controls like "Run", "Share", "Schedule", "Open in", "More", "Save query", and "Download".

Q4. DISTINCT

The screenshot shows the Google Cloud BigQuery console interface. The query editor displays the following SQL code:

```
16 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`;  
17  
18 --Q4  
19 SELECT DISTINCT(`Product Category`) AS product_category  
20 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`;  
21
```

Below the query editor, a status bar indicates "Using on-demand processing quota" and a message: "This script will process 342.34 KB when run." The "Query results" section is active, showing a table with three rows of data:

Row	product_category
1	Beauty
2	Clothing
3	Electronics

The interface includes standard BigQuery controls like "Run", "Share", "Schedule", "Open in", "More", "Save query", and "Download".

Q5. GROUP BY

🏠 X 🔍 Untitled...ery X 🔍 *Practic... nal X + ▾

🔍 Practical3 Final ▶ Run 👤 Share ▾ 🕒 Schedule 🔗 Open in ▾ ⚙️ More ▾ 💾 Save query ▾ ⬇️ Download

```
23 --Q5
24 SELECT `Product Category` AS product_category,
25 SUM(Quantity) AS total_quantity
26 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`
27 GROUP BY `Product Category`;
28
```

✔️ This script will process 342.34 KB when run.
Using on-demand processing quota

Query results Chat Save results ▾ Open in ▾ ↕

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	product_category ▾	total_quantity ▾
1	Beauty	771
2	Clothing	894
3	Electronics	849

Q6. CASE statement

Untitled...ery ×

*Practic... nal ×

Practical3 Final

Run

Share

Schedule

Open in

More

Save query

Download

```
--Q6
30 SELECT `Transaction ID` AS customer_ID,
31       Age,
32       CASE
33         WHEN Age < 30 THEN 'Youth'
34         WHEN Age BETWEEN 30 AND 59 THEN 'Youth'
35         ELSE 'Senior'
36       END AS Age_Group
37 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`;
38
39
```

✔ This script will process 342.34 KB when run.

Using on-demand processing quota

Query results

ChatSave resultsOpen in

Job informationResultsVisualizationJSONExecution detailsExecution graph

Row	customer_ID	Age	Age_Group
1	191	64	Senior
2	204	39	Youth
3	230	54	Youth
4	232	43	Youth
5	309	26	Youth
6	310	28	Youth
7	363	64	Senior
8	371	20	Youth
9	397	30	Youth
10	454	46	Youth
11	512	57	Youth
12	739	36	Youth
13	791	51	Youth
14	825	46	Youth
15	855	54	Youth
16	945	30	Youth
17	967	62	Senior
18	979	19	Youth

Results per page: 501 – 50 of 1000

Q 7. Conditional Aggregation

Untitled...ery X

*Practic... nal X

+

Practical3 Final

Run

Share

Schedule

Open in

More

Save query

Download

```
39
40 --Q7
41 SELECT Gender,
42 COUNT('Transaction ID') As high_value_transactions
43 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`
44 WHERE `Total Amount` > 500
45 GROUP BY Gender;
46
```

✓ This script will process 350.15 KB when run.

Using on-demand processing quota

Query results

Chat Save results Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

Q8. HAVING Clause

Untitled...ery X *Practic... nal X

Practical3 Final Run Share Schedule Open in More Save query Download

```
40
49 --Q8
50 SELECT `Product Category` AS product_category,
51 SUM(`Total Amount`) AS total_revenue
52 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`
53 GROUP BY `Product Category`
54 HAVING total_revenue > 5000;|
55
```

✓ This script will process 350.15 KB when run.

Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	product_category	total_revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Q9. Calculated fields

Untitled...ery X *Practic... nal X

Practical3 Final Run Share Schedule Open in More Save query Download

```
57
58 --Q9
59 SELECT `Transaction ID` AS transaction_ID,
60        `Price per Unit` AS price_per_unit,
61        CASE
62          WHEN `Price per Unit` < 50 THEN 'Cheap'
63          WHEN `Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
64          ELSE 'Expensive'
65        END AS unit_cost_category
66 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`;
67
68
```

✔ This script will process 350.15 KB when run.

Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	transaction_ID	price_per_unit	unit_cost_category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap
7	363	25	Cheap
8	371	25	Cheap
9	397	25	Cheap
10	454	25	Cheap
11	512	25	Cheap
12	739	25	Cheap
13	791	25	Cheap
14	825	25	Cheap
15	855	25	Cheap
16	945	25	Cheap
17	967	25	Cheap
18	970	25	Cheap

Results per page: 50 1 – 50 of 1000 |< < > >|

Q10. Combining WHERE + CASE

Untitled...ery × *Practic... nal ×

Practical3 Final Run Share Schedule Open in More Save query Download

```
70 ---Q10
71 SELECT `Customer ID` AS customer_ID,
72        Age,
73        `Total Amount` AS total_amount,
74        CASE
75          WHEN `Total Amount` > 1000 THEN 'High'
76          ELSE 'Low'
77        END AS spending_level
78 FROM `bigquery-practical-494621.bigquerypractical.retailpractical3`
79 WHERE Age > 40;
80
81
82
```

✓ This script will process 350.15 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	customer_ID	Age	total_amount	spending_level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low
5	CUST454	46	25	Low
6	CUST512	57	25	Low
7	CUST791	51	25	Low
8	CUST825	46	25	Low
9	CUST855	54	25	Low
10	CUST967	62	25	Low
11	CUST006	45	30	Low
12	CUST057	63	30	Low
13	CUST344	42	30	Low
14	CUST376	64	30	Low
15	CUST387	44	30	Low
16	CUST396	55	30	Low
17	CUST749	42	30	Low

Results per page: 50 1 – 50 of 534